

Coherent background

B and $B + \sqrt{B_i} e_i$
One controlled bin change
for each support bin



Run the full analysis

Baseline + response scans
Obtain offset a and response D
Keep the complete mass grid



Build the field covariance

$$\begin{aligned}\Gamma &= D^T D \\ s_m^2 &= \Gamma_{mm} \\ R_{mn} &= \Gamma_{mn} / (s_m s_n)\end{aligned}$$



Sample entire fields

$z^* \sim N(0, R)$
 $r^* = a + s \odot z^*$
Evaluate one maximum per field



Independent validation

Complete Poisson spectra run through the original analysis.
Check distributions, correlations and accessible tails.